WORKED SOLUTION

STEP III 2006, Question 9

Mechanics • Energy

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Read the question before you go on. It is in the STEP Support Programme's question database:

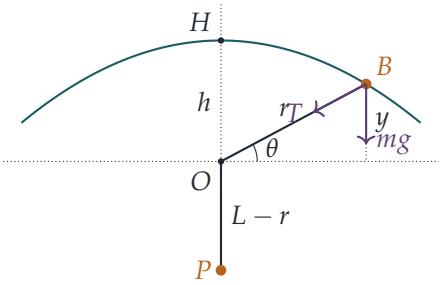
step.maths.org/questions/06-s3-q9

The key idea

The string has fixed length, so whatever the bead gains in distance from O, the particle P rises by the same amount. That ties the two potential energies together: the total depends only on $y + r$. "Equilibrium in every position" means the potential energy has no preferred position, so it must be constant along the wire. Once it is constant, energy conservation says the total kinetic energy never changes, and the rest is writing the bead's speed in polar coordinates.

Solution

Model. P and B are particles of mass m; the string is light and inextensible; the ring and the wire are smooth, so the string tension does no net work and the wire's normal reaction does no work. Take O as origin, with x horizontal and y vertically up, and let the string have length L.



The shape of the wire

The length of string from O to B is r, so the length OP is $L - r$ and P is a distance $L - r$ below O. Taking the horizontal plane through O as the zero of potential energy, the

total potential energy is

$$U = mgy - mg(L - r) = mg(y + r) - mgL.$$

The system has one degree of freedom (the position of the bead on the wire), and an equilibrium position is a stationary point of U along the wire. If every position is an equilibrium, then $\mathrm{d}U/\mathrm{d}s = 0$ everywhere along the wire (s being arc length), so U is constant along the wire. Hence $y + r$ is constant.

At H the bead is directly above O at height h, so $y = h$ and $r = h$ there, and the constant is $2h$. So the wire is

$$y + r = 2h,$$

as required. (In Cartesian form, $x^2 + y^2 = (2h - y)^2$ gives $y = h - x^2/(4h)$: a parabola with focus O, as in the picture. You can also confirm the equilibrium directly: the tension is mg, since P is in equilibrium, and the resultant of mg along BO and mg downwards bisects the angle between those directions, which is exactly the normal to a parabola at a point, by its reflection property.)

The rate of change of θ

On the wire $y = 2h - r$, so

$$\sin\theta = \frac{y}{r} = \frac{2h}{r} - 1.$$

Differentiating with respect to time,

$$\cos\theta\,\dot{\theta} = -\frac{2h\dot{r}}{r^2}.$$

Since $\theta = \arcsin(y/r)$ lies in $[-\frac{\pi}{2}, \frac{\pi}{2}]$, we have $\cos\theta \geq 0$, so

$$\cos\theta = \sqrt{1 - \left(\frac{2h - r}{r}\right)^2} = \frac{\sqrt{r^2 - (2h - r)^2}}{r} = \frac{\sqrt{4rh - 4h^2}}{r} = \frac{2\sqrt{rh - h^2}}{r}.$$

After the bead leaves H we have $r > h$ (because $r = 2h - y$ and $y < h$ away from H), so $\cos\theta \neq 0$ and we may divide:

$$\dot{\theta} = -\frac{2h\dot{r}}{r^2} \cdot \frac{r}{2\sqrt{rh - h^2}} = -\frac{h\dot{r}}{r\sqrt{rh - h^2}},$$

which is the required result.

The speed of P

Ask yourself Why must the total kinetic energy stay equal to $\frac{1}{2}mV^2$?

The only forces that do work on the system are gravity and the (internal) string tension. The tension does no net work: the string is inextensible and the ring smooth, so the work done by the tension on B cancels that done on P. The wire is smooth, so its

reaction does no work. Hence kinetic plus potential energy is conserved. But we have just shown that the potential energy U is *constant* along this wire, so the kinetic energy is constant too.

At the start the bead is at H, where r has its minimum value h (since y is greatest there). So $\dot{r} = 0$ initially, and P, whose speed is $|\dot{r}|$, starts at rest. The initial kinetic energy is therefore $\frac{1}{2}mV^2$.

In polar coordinates about O, the bead's speed squared is $\dot{r}^2 + r^2\dot{\theta}^2$. Using the result for $\dot{\theta}$,

$$r^2\dot{\theta}^2 = \frac{h^2\dot{r}^2}{rh - h^2} = \frac{h\dot{r}^2}{r-h}, \qquad \text{so} \qquad v_B^2 = \dot{r}^2\left(1 + \frac{h}{r-h}\right) = \frac{r\dot{r}^2}{r-h}.$$

Conservation of kinetic energy, after cancelling $\frac{1}{2}m$, gives

$$\frac{r\dot{r}^2}{r-h} + \dot{r}^2 = V^2 \quad \Longrightarrow \quad \dot{r}^2\,\frac{2r-h}{r-h} = V^2.$$

Since $r > h$, both $r - h$ and $2r - h$ are positive, and the speed of P is

$$|\dot{r}| = V\left(\frac{r-h}{2r-h}\right)^{\frac{1}{2}},$$

as required.

Checks

- At H, $r = h$ and the formula gives 0: correct, since P starts at rest.
- As $r \to \infty$ the speed of P tends to $V/\sqrt{2}$. Far out the bead moves almost radially, so B and P have nearly equal speeds and share the kinetic energy equally.
- Numerically, with $h = 1$ and $V = 2$, we stepped the bead's equation of motion along $y = 1 - \frac{1}{4}x^2$ (with RK4) and compared finite-difference values of $\dot{\theta}$ and $\dot{r}$ with the formulae. At $r \approx 4.043$ the speed of P was 1.31064, matching $2\sqrt{3.043/7.087}$ to six figures.

Common slips

- Taking the potential energy of P as $+mg(L - r)$: P is *below* O, so its height is $-(L - r)$.
- Forgetting that P moves too, and writing $\frac{1}{2}mv_B^2 = \frac{1}{2}mV^2$.
- Dividing by $\cos\theta$ without noting that it vanishes only at H.

Beyond STEP

Finding a curve on which a system is in "neutral" equilibrium everywhere is a small instance of the principle that equilibria are stationary points of the potential energy,

which leads on to Lagrangian mechanics, where constraints like the string and wire are built in by choosing good coordinates. Here the single coordinate r does the job, and energy conservation is the first integral. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers energy methods and polar coordinates in this spirit.

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